AP Calculus II
Notes 8.1
Basic Integration Techniques

In this section, we will review all the form-fitting integration techniques we have learned to get a firm grasp on when to use; which to not only get them correct, but to do so quickly and efficiently.

<u>Ex. 1:</u> Evaluate the following closely related integrals:

a) $\int \frac{4}{x^2+9}\,dx$

$u^2 = x^2$ $\quad a^2 = 9$

$u = x$ $\quad a = 3$

$du = 1\,dx$

arctan!!

$\boxed{\frac{1}{3}\arctan\frac{x}{3} + C}$

b) $\int \frac{4x}{x^2+9}\,dx$

$u = x^2 + 9$

$du = 2x\,dx$

$\frac{4}{2}\int \frac{2x}{x^2+9}\,dx$

$= \boxed{2\ln|x^2+9| + C}$

c) $\int \frac{4x^2}{x^2+9}\,dx$

long division!

$$\begin{array}{r} 4 \\ x^2+9 \enclose{longdiv}{4x^2} \\ -\ \ 4x^2+36 \\ \hline -36 \end{array}$$

$\int \left(4 + \frac{-36}{x^2+9}\right)dx$

$= \int 4\,dx - 36\int \frac{1}{x^2+9}\,dx$

arctan!!

$= 4x - 36 \cdot \frac{1}{3}\arctan\frac{x}{3} + C$

$= \boxed{4x - 12\arctan\frac{x}{3} + C}$

It is VERY important to know all the integration rules inside and out so you know which integrals apply to the certain rules. Do NOT force a problem into a rule; it must fit perfectly.

Ex. 2: Evaluate $\int_0^1 \frac{x+3}{\sqrt{4-x^2}}\,dx$. $\xrightarrow[\text{up!}]{\text{split}}$ $\int_0^1 \frac{x}{\sqrt{4-x^2}}\,dx + 3\int_0^1 \frac{1}{\sqrt{4-x^2}}\,dx$

$-\frac{1}{2}\int_0^1 \frac{-2x}{\sqrt{4-x^2}}\,dx$

$u = 4-x^2$ $\quad x=0 \to u=4$

$du = -2x\,dx$ $\quad x=1 \to u=3$

$-\frac{1}{2}\int_4^3 \frac{1}{\sqrt{u}}\,du = \frac{1}{2}\int_3^4 u^{-1/2}\,du$

$= \frac{1}{2}\cdot\frac{u^{1/2}}{1/2}\Big|_3^4 = \sqrt{u}\Big|_3^4$

$+3\int_0^1 \frac{1}{\sqrt{4-x^2}}\,dx$ $\quad u^2 = x^2$, $\boxed{u=x}$, $du = dx$

$+3\int_0^1 \frac{1}{\sqrt{a^2-u^2}}\,du$ $\quad a^2=4,\ a=2$; $x=0 \to u=0$; $x=1 \to u=1$

$+3\arcsin\frac{x}{2}\Big|_0^1$

$+(3\arcsin\frac{1}{2} - 3\arcsin 0)$

$= (2-\sqrt{3}) + (\frac{3\pi}{6} - 0) = \boxed{2-\sqrt{3}+\frac{\pi}{2}}$

Ex. 3: Find the value of k such that the area under the graph of $f(x) = \frac{e^x}{1+e^x}$ from $x = 0$ to $x = k$ is $\ln 3$.

integrate!

$\text{Area} = \int_0^k \frac{e^x}{1+e^x}\,dx = \ln 3$

to get k out of bounds, integrate & use FTC!

$u = 1+e^x$ $\quad x=0 \to u=2$

$du = e^x\,dx$ $\quad x=k \to u=1+e^k$

$= \int_2^{1+e^k} \frac{du}{u} = \ln 3 \longrightarrow \ln|u|\Big|_2^{1+e^k} = \ln 3$

$= \ln|1+e^k| - \ln 2 = \ln 3, \quad e^{\ln\left|\frac{1+e^k}{2}\right|} = e^{\ln 3}$

$\frac{1+e^k}{2} = 3, \quad 1+e^k = 6, \quad e^k = 5, \quad \boxed{k = \ln 5}$

Ex. 4: The rate in which snow is falling can be modeled by the equation $R(t) = \frac{\cos^2 t+\sin t}{\cos t}$ for $t \geq 0$. How much snow has fallen on the ground from $0 \leq t \leq \frac{\pi}{4}$?

$$\int_0^{\pi/4} \frac{\cos^2 t+\sin t}{\cos t}\,dt \rightarrow \int_0^{\pi/4}(\cos t+\tan t)\,dt$$

$$= \sin t - \ln|\cos t|\Big|_0^{\pi/4}$$

$$= \left(\tfrac{1}{\sqrt{2}} - \ln\left|\tfrac{1}{\sqrt{2}}\right|\right) - (0 - \cancel{\ln 1})$$

$$= \tfrac{1}{\sqrt{2}} - \ln\tfrac{1}{\sqrt{2}} = \boxed{\tfrac{1}{\sqrt{2}} + \ln\sqrt{2}}$$

Ex. 5: A particle moves along the x −axis such that its velocity can be modeled by the function $v = t\left(1+\frac{1}{t}\right)^2$ for $t > 0$. Find the position of the particle at $t = 3$ given that $x(1) = 4$.

net change!!!

$$X(3) = 4 + \int_1^3 t\left(1+\tfrac{1}{t}\right)^2 dt$$

$u = 1 + 1/t$
$du = -1/t^2\, dt$
NOPE!!

$$x(3) = 4 + \int_1^3 t\left(1+\tfrac{2}{t}+\tfrac{1}{t^2}\right)dt$$

$$x(3) = 4 + \int_1^3 \left(t+2+\tfrac{1}{t}\right)dt$$

$$= 4 + \left[\tfrac{1}{2}t^2+2t+\ln|t|\Big|_1^3\right]$$

$$= 4 + \left(\tfrac{9}{2}+6+\ln 3\right) - \left(\tfrac{1}{2}+2+\cancel{\ln 1}\right)$$

$$= 4+4+4+\ln 3 = \boxed{12+\ln 3}$$

Summary of Procedures to Fit Integrands Into Integration Rules

Ex. $(1+e^x)^2$

1) Multiply/Expand!

$= 1 + 2e^x + e^{2x}$

Ex. $\frac{1+x}{x^2+1} = \frac{1}{x^2+1} + \frac{x}{x^2+1}$

2) Separate numerator!

Ex. $\frac{3}{5+2x+x^2} = \frac{3}{x^2+2x+5}$

3) complete the square!

$\frac{2}{2} = 1$

$(1)^2 = 1$

$= \frac{3}{x^2+2x+1+5-1} = \frac{3}{(x+1)^2+4}$

Ex. $\frac{x^2}{x^2+1}$

4) Long Division!

$$\begin{array}{r} 1 \\ x^2+1 \overline{)\ x^2 \quad} \\ -\ (x^2+1) \\ \hline -1 \end{array} = 1 + \frac{-1}{x^2+1}$$

Ex. cot^2x

5) Trig Identity!!

$1 + \cot^2 x = \csc^2 x$

$= \csc^2 x - 1$

AP Calculus II
Notes 8.2
Integration by Parts

In this section, we now look at a new technique that is very useful for integrands involving products of algebraic and transcendental functions. Integration by parts is based on the formula for the derivative of a product:

$$\frac{d}{dx}[uv] = u\frac{dv}{dx} + v\frac{du}{dx}$$

If we integrate both sides of this equation, we get:

$$\int \frac{d}{dx}[uv]\,dx = \int u\frac{dv}{dx}\,dx + \int v\frac{du}{dx}\,dx$$

By rewriting this equation, you obtain the following integration rule: $uv = \int u\,dv + \int v\,du$

$$\boxed{\int u\,dv = uv - \int v\,du}$$

Theorem – Integration by Parts

If u and v are functions of x and have continuous derivatives, then

$$\int u \cdot dv = uv - \int v\,du$$

product of 2 different functions

* note we need $u \to du$ and $dv \to v$

* there's an integral after doing this *

Guidelines for Integration by Parts:

1) Try letting dv be the most complicated portion of the integrand that fits a basic integration rule.
2) Try letting u be the portion of the integrand whose derivative is a function **simpler** than u.
3) The dv always includes the dx of the original integrand.

A little help for picking which part is u …..

Inverse trig (arcsin, arctan)

Ln

Algebraic expressions ($x, x^2, x+9\ldots$)

Trig

E^x

Why?? Since $u \to du$, we want functions whose derivative makes it easier to work with… like inverse trig & ln … trig remains trig & e^x stays e^x

Ex. 1: Integrate the following using Integration by Parts when necessary:

a) $\int 3x^2 \ln x \, dx$

$u = \ln x$, $du = \frac{1}{x}dx$ ... doesn't work

since a product, let's try integration by parts!

$\int \underline{3x^2} \, \underline{\ln x} \, \underline{dx}$

$u = \ln x$ $\quad \int dv = \int 3x^2 \, dx$

$du = \frac{1}{x}dx$ $\quad v = x^3$

note we will get $+C$ at end!

$\int u \, dv = uv - \int v \, du$

$\int 3x^2 \ln x \, dx = (\ln x)(x^3) - \int x^3 \cdot \frac{1}{x} dx$

if we picked the right combo, the subsequent integral should be simpler

$= x^3 \ln x - \int x^2 \, dx$ yeah... that's simpler!

$= \boxed{x^3 \ln x - \frac{1}{3}x^3 + C}$

b) $\int x e^{\frac{2x}{3}} dx$

$u = \frac{2x}{3}$, $du = \frac{2}{3}dx$ .... nope, let's try integration by parts!

ILATE says $u = x$ (A before E)

$u = x$ $\quad \int dv = \int e^{2/3 x} dx$ ← this requires a u-sub! $u = \frac{2}{3}x$ $\; du = \frac{2}{3}dx$

$du = 1 \, dx$ $\quad v = \frac{3}{2} e^{2/3 x}$

$$\int x e^{\frac{2}{3}x} dx = (x)\left(\frac{3}{2} e^{2/3 x}\right) - \int \left(\frac{3}{2} e^{2/3 x}\right)(dx)$$

$$= \frac{3}{2} x e^{2/3 x} - \frac{3}{2} \cdot \frac{3}{2} \int \frac{2}{3} e^{2/3 x} dx$$

← this is simpler! $u = \frac{2}{3}x$ $\; du = \frac{2}{3}dx$

$$= \boxed{\frac{3}{2} x e^{\frac{2}{3}x} - \frac{9}{4} e^{2/3 x} + C}$$

Ex. 3: The acceleration of a particle moving along the x −axis is modeled by $a(t) = t^2 \cos t$ for all $t \geq 0$. Write the velocity equation for any time t given $v(0) = -2$.

$a(t) \to v(t)$

integrate!

$$v(t) = -2 + \int_0^t a(x)\,dx$$ or indefinite integral!

$$v(t) = \int t^2 \cos t\,dt \longrightarrow v(t) = t^2 \sin t - 2\int t \sin t\,dt$$

$u = t^2$ $\quad \int dv = \int \cos t\,dt$

$du = 2t\,dt$ $\quad v = \sin t$

simpler!!

But we need another round!!

$$v(t) = t^2 \sin t - 2\int t \sin t\,dt \longrightarrow t^2 \sin t - 2\left[-t\cos t + \int \cos t\,dt\right]$$

simpler!!

$u = t,\ du = dt \ /\ \int dv = \int \sin t\,dt,\ v = -\cos t$

$$v(t) = t^2 \sin t + 2t\cos t - 2\sin t + C \xrightarrow[C=-2]{v(0)=-2} \boxed{v(t) = t^2 \sin t + 2t\cos t - 2\sin t - 2}$$

Ex. 3: Find the area of the region bounded by the x −axis, the graph $y = \arcsin x$ and $x = 0$ and $x = \frac{1}{2}$.

$$\int_0^{1/2} \arcsin x\,dx$$

hidden integration by parts!!

ILATE says $u = \arcsin x$!!

simpler!

$u = \arcsin x$ $\quad \int dv = \int dx$

$du = \frac{1}{\sqrt{1-x^2}}\,dx$ $\quad v = x$

$$\longrightarrow x \arcsin x \Big|_0^{1/2} - \int_0^{1/2} \frac{x}{\sqrt{1-x^2}}\,dx$$

part of antiderivative, so needs evaluating!

$$\text{Area} = x\arcsin x\Big|_0^{1/2} - \frac{-1}{2}\int_0^{1/2} \frac{-2x}{\sqrt{1-x^2}}\,dx$$

$u = 1 - x^2$ $\quad x = 0 \to u = 1$

$du = -2x\,dx$ $\quad x = 1/2 \to u = 3/4$

$$= x\arcsin x\Big|_0^{1/2} + \frac{1}{2}\int_1^{3/4} u^{-1/2}\,du \longleftarrow -\frac{1}{2} \cdot \frac{u^{1/2}}{1/2} = -\sqrt{u}$$

$$= x\arcsin x\Big|_0^{1/2} - \sqrt{u}\Big|_{3/4}^{1} = \frac{1}{2}\arcsin\frac{1}{2} - 0 - \left(1 - \sqrt{3/4}\right)$$

$$= \frac{1}{2}\cdot\frac{\pi}{6} - 1 + \frac{\sqrt{3}}{2} = \boxed{\frac{\pi}{12} - 1 + \frac{\sqrt{3}}{2}}$$

Ex. 4: Let f be a differentiable function such that $\int f(x)\sin x\,dx = -f(x)\cos x + \int 4x^3\cos x\,dx$. Which of the following could be $f(x)$?

(A) $\sin x$ (B) $4x^3$ (C) $-x^4$ (D) x^4

$$\int u\,dv = uv - \int v\,du$$

v appears to be trig

$$\int f(x)\sin x\,dx = -f(x)\cos x + \int 4x^3\cos x\,dx$$

$u = f(x)$ $dv = \sin x\,dx$

$du = f'(x)\,dx$ $v = -\cos x$ $\rightarrow -f(x)\cos x + \int f'(x)\cos x\,dx$

so $f'(x) = 4x^3$, $f(x) = x^4$

Ex. 5:

x	1	2	3	4	5
$f'(x)$	62	30	20	15	12

The function f is twice differentiable for $x \geq 1$ where $f(5) = -6$. Selected values of the positive and decreasing function f', the derivative of f, are given in the table above.

If $\int_1^5 xf''(x)dx = -100$, find $f(1)$. integration by parts!

we don't know if f'' is trig/e^x/ln, etc.... but we need to get to f... so let f'' go to f

$u = x$ $dv = f''(x)\,dx$

$du = 1\,dx$ $v = f'(x)$

simpler!!

$$\int_1^5 xf''(x)dx = xf'(x)\Big|_1^5 - \int_1^5 f'(x)dx$$

$$-100 = 5f'(5) - f'(1) - \left[f(x)\Big|_1^5\right]$$

$$-100 = 5(12) - 62 - f(5) + f(1)$$

$-100 = -2 + 6 + f(1)$, $f(1) + 4 = -100$, $\boxed{f(1) = -104}$

And now, the moment you've all been waiting for, Mr. Gough's favorite problem in all of calculus!

<u>Ex. 6:</u> Evaluate $\int e^x \sin x \, dx$ ILATE says $u = \sin x$

$u = \sin x \qquad dv = e^x dx$

$du = \cos x \, dx \qquad v = e^x$

$$\int e^x \sin x \, dx = e^x \sin x - \underbrace{\int e^x \cos x \, dx}$$

another round!!

$u = \cos x \qquad dv = e^x dx$

$du = -\sin x \, dx \qquad v = e^x$

$$\int e^x \sin x \, dx = e^x \sin x - \left[e^x \cos x + \int e^x \sin x \, dx \right]$$

$$\int e^x \sin x \, dx = e^x \sin x - e^x \cos x - \underbrace{\int e^x \sin x \, dx}$$

another round!!!

wait ... this will bring us back to the beginning... infinite rounds??? what if we got creative

If I'm trying to solve for $\int e^x \sin x \, dx$, let's view this as an algebraic equation & add the integral to both sides!

$$2 \int e^x \sin x \, dx = e^x \sin x - e^x \cos x + C$$

$$\int e^x \sin x \, dx = \boxed{\frac{1}{2}\left(e^x \sin x - e^x \cos x\right) + C}$$

we need itself to answer itself ....

wow

Review:

$$\frac{3}{x+2}+\frac{4}{x+5}=\frac{3(x+5)+4(x+2)}{(x+2)(x+5)}=\frac{3x+15+4x+8}{x^2+7x+10}=\boxed{\frac{7x+23}{x^2+7x+10}}$$

AP Calculus II
Notes 8.5
Partial Fractions

This section takes a look at another integration technique in which a rational function is decomposed into simpler rational functions to make the integration process easier. Before we actually integrate, let's check out how to decompose rational functions.

Decomposition goes backwards!!

Ex. 1: Write the partial fraction decomposition for $\frac{4x+7}{x^2+x-6}$.

$$\frac{4x+7}{(x+3)(x-2)}$$

← so there are 2 fractions that when added become this. The denominators are each factor, but we don't know the constants that will result in $4x+7$ up top.

$$\frac{A}{x+3}+\frac{B}{x-2}:\quad \frac{4x+7}{(x+3)(x-2)}=\frac{A}{x+3}+\frac{B}{x-2}$$

multiply both sides by $(x+3)(x-2)$!

plug in x-values to find A & B. Smart x-values would be the zeros!

$$4x+7=A(x-2)+B(x+3)$$

$x=2$: $15=\cancel{0A}+5B$, $\boxed{B=3}$

$x=-3$: $-5=-5A+\cancel{0B}$, $\boxed{A=1}$

$$\Rightarrow \boxed{\frac{1}{x+3}+\frac{3}{x-2}}$$

Ex. 2: Write the partial fraction decomposition for $\frac{5x^2+20x+}{x^3+3x^2+2x}$. $(x^3+3x^2+2x) \to x(x+1)(x+2)$

$$\frac{5x^2+20x+6}{x(x+1)(x+2)}=\frac{A}{x}+\frac{B}{x+1}+\frac{C}{x+2}$$

* jump straight to part without fraction *

$$5x^2+20x+6=A(x+1)(x+2)+Bx(x+2)+Cx(x+1)$$

$x=-1$: $-9=-B$, $B=9$

$x=-2$: $-14=2C$, $C=-7$

$x=0$: $6=2A$, $A=3$

$$\boxed{\frac{5x^2+20x+6}{x^3+3x^2+2x}=\frac{3}{x}+\frac{9}{x+1}-\frac{7}{x+2}}$$

can't integrate

much easier to integrate

Ex. 3: Evaluate $\int \frac{3x-1}{x^2-x-6}\,dx$.

$u = x^2 - x - 6$

$du = (2x-1)\,dx \ \ldots$ off by more than just a constant

since you can factor the denominator, use partial fractions!!

$$\frac{3x-1}{x^2-x-6} = \frac{A}{x-3} + \frac{B}{x+2} \Rightarrow 3x-1 = A(x+2) + B(x-3)$$

$\underline{x=-2}$

$-7 = -5B$

$B = 7/5$

$\underline{x=3}$

$8 = 5A$

$A = 8/5$

$$\text{So, } \int \frac{3x-1}{x^2-x-6}\,dx = \int\left(\frac{8/5}{x-3} + \frac{7/5}{x+2}\right)dx$$

$$= \frac{8}{5}\int \frac{1}{x-3}\,dx + \frac{7}{5}\int \frac{1}{x+2}\,dx = \boxed{\frac{8}{5}\ln|x-3| + \frac{7}{5}\ln|x+2| + C}$$

$u = x-3$ $\quad \int \frac{du}{u}$

$du = 1\,dx$

$u = x+2$ $\quad \int \frac{du}{u}$

$du = 1\,dx$

Ex. 4: Evaluate $\int \frac{21}{2x^2+x-6}\,dx$

$$\frac{21}{2x^2+x-6} = \frac{A}{2x-3} + \frac{B}{x+2}$$

$$21 = A(x+2) + B(2x-3)$$

$\underline{x=-2}$

$21 = -7B$

$B = -3$

$\underline{x = 3/2}$

$21 = 7/2\,A$

$A = 6$

(or if you don't like fractions)

$\underline{x=0}$ ← any other constant

$21 = 2A - 3B$

$21 = 2A - 3(-3)$

$21 = 2A + 9$

$12 = 2A$, $\underline{A=6}$

$$\int \frac{21}{2x^2+x-6}\,dx = 6\int \frac{1}{2x-3}\,dx - 3\int \frac{1}{x+2}\,dx$$

$u = 2x-3$

$du = 2\,dx$

$u = x+2$

$du = 1\,dx$

$$= \frac{6}{2}\int \frac{du}{u} - 3\int \frac{du}{u} = \boxed{\begin{aligned} &3\ln|2x-3| - 3\ln|x+2| + C \\ &= 3\ln\left|\frac{2x-3}{x+2}\right| + C \end{aligned}}$$

Ex. 5: The population of Haverford is growing at a rate modeled by $G(t) = \frac{t^2+3t-30}{t^2-25}$ where $G(t)$ is millions of people per year and t is years since 2010 ($t = 0$). If there are 123 million people in the year 2011, what will the population be in 2014? $\to t=4$

$\hookrightarrow t=1$

$$123 + \int_1^4 \frac{t^2+3t-30}{t^2-25}\,dt$$

← must do long division first!

$$t^2-25 \overline{\big)\, t^2+3t-30}\quad 1 + \frac{3t-5}{t^2-25}$$
$$-\;(t^2 \quad -25)$$
$$3t-5$$

$$123 + \int_1^4 1\,dt + \int_1^4 \frac{1}{t^2-25}\,dt$$

$u=t^2-25$
$du=2t\,dt$ ← Nope.... But is factorable!!

$$\frac{3t-5}{t^2-25} = \frac{A}{t-5} + \frac{B}{t+5}$$

$$3t-5 = A(t+5) + B(t-5)$$

$t=-5$	$t=5$
$-20=-10B$	$10=10A$
$B=2$	$A=1$

$$123 + \int_1^4 1\,dt + \int_1^4 \frac{1}{t-5}\,dt + 2\int_1^4 \frac{1}{t+5}\,dt$$

$u=t-5$, $du=1dt$ $\qquad$ $u=t+5$, $du=1dt$

$$123 + t\Big|_1^4 + \ln|u|\Big|_{-4}^{-1} + 2\ln|u|\Big|_6^9$$

$$=123+(4-1)+\ln 1-\ln 4+2\ln 9-2\ln 6$$

$$=126+\ln\left|\frac{9^2}{6^2\cdot 4}\right| = \boxed{126+\ln|9/16|}$$

Ex. 6: $\int \frac{12}{(x-1)(x-5)}\,dx =$

(A) $-3\ln|x-1| + 3\ln|x-5| + C$

(B) $-2\ln|x-1| + 2\ln|x-5| + C$

(C) $3\ln|x-1| - 3\ln|x-5| + C$

(D) $12\ln|x-1| + 12\ln|x-5| + C$

$$\frac{12}{(x-1)(x-5)} = \frac{A}{x-1} + \frac{B}{x-5}$$

$$12 = A(x-5) + B(x-1)$$

$x=5$	$x=1$
$12=4B$	$12=-4A$
$B=3$	$A=-3$

$$3\int \frac{1}{x-5}\,dx + -3\int \frac{1}{x-1}\,dx$$

$u=x-5$, $du=1dx$ $\qquad$ $u=x-1$, $du=1dx$

$$\boxed{3\ln|x-5| - 3\ln|x-1| + C}$$

Ex. 7: Evaluate $\int \frac{2x^2-6x-18}{x^3-9x}dx$

$\frac{2x^2-6x-18}{x^3-9x} = \frac{A}{x} + \frac{B}{x+3} + \frac{C}{x-3}$

$u = x^3-9x$, $du = 3x^2-9\,dx$ nope!

$2x^2-6x-18 = A(x-3)(x+3) + Bx(x-3) + Cx(x+3)$

$\underline{x=3}$
$-18 = 18C$
$C = -1$

$\underline{x=-3}$
$18 = 18B$
$B = 1$

$\underline{x=0}$
$-18 = -9A$
$A = 2$

$= 2\int \frac{1}{x}dx + \int \frac{1}{x+3}dx - \int \frac{1}{x-3}dx$

$u = x+3$
$du = dx$

$u = x-3$
$du = 1dx$

$= \boxed{2\ln|x| + \ln|x+3| - \ln|x-3| = \ln\left|\frac{x^2(x+3)}{x-3}\right| + C}$

Ex. 8: Describe what process and which formula you would use, if any, to integrate.

a) $\int \frac{4}{x^2+10x+30}dx$

complete the square

b) $\int \frac{4x+6}{x^2+3x-1}dx$

U-sub

c) $\int \frac{x^2+9x-15}{2x}dx$

separate numerator

d) $\int \frac{4}{x^2+10x+24}dx$

partial fractions

e) $\int \frac{4x^3+31}{x^2+10}dx$

long division then arctan

f) $\int \frac{4\cos x}{\sin^2 x + 6\sin x + 8}dx$

partial fractions!

[Try $u = \sin x$....]

$\int \frac{4}{u^2+6u+8}du$

AP Calculus II
Notes 8.7
Indeterminate Forms/L'Hôpital's Rule

<u>Review of Limits</u>:

Evaluate the following limits:

a) $\lim_{x\to 4} \frac{x^2-x-12}{x^2-16} \to \frac{0}{0}$

$\lim_{x\to 4} \frac{\cancel{(x-4)}(x+3)}{\cancel{(x-4)}(x+4)} = \boxed{\frac{7}{8}}$

b) $\lim_{x\to\infty} \frac{3x^2-x-12}{4x^2-16} \cdot \frac{1/x^2}{1/x^2}$

$\lim_{x\to\infty} \frac{3-1/x-12/x^2}{4-16/x^2} = \frac{3-0-0}{4-0} = \boxed{\frac{3}{4}}$

c) $\lim_{t\to\frac{\pi}{2}} \frac{1-\sin^2 t}{t\cos t} \to \frac{0}{0}$

$\lim_{t\to\pi/2} \frac{\cos^2 t}{t\cos t}$

$\lim_{t\to\pi/2} \left(\frac{\cos t}{t}\right) = \frac{0}{\pi/2} = \boxed{0}$

d) $\lim_{h\to 0} \frac{\cos(x+h)-\cos x}{h}$

definition of derivative of cos x

$\boxed{-\sin x}$

Now, what about $\lim_{x\to 0} \frac{e^x-1}{\sin x}$? If you can't figure out a way to simplify this, then use the table/graph to evaluate the limit.

$\lim_{x\to 0} \frac{e^x-1}{\sin x} =$ ← can't factor, no known identities ....

L'Hôpital's Rule:

Let f and g be functions that are differentiable on an open interval (a, b) with c, except possibly at c.

If the $\lim_{x\to c} f(x) = 0$ and $\lim_{x\to c} g(x) = 0$, then $\lim_{x\to c} \frac{f(x)}{g(x)}$ is indeterminate and $\lim_{x\to c} \frac{f(x)}{g(x)} = \lim_{x\to c} \frac{f'(x)}{g'(x)}$

This result also applies if the $\lim_{x\to c} \frac{f(x)}{g(x)}$ is some other indeterminate form $\frac{\pm\infty}{\pm\infty}, \pm\infty \mp \infty, 0 \bullet \infty$, etc…

There are three major mistakes that are made in L'Hôpital's Rule:

a) Not checking if indeterminate or writing "$=\frac{0}{0}$"

b) Doing a quotient rule

c) Poor notation.... dropping limit notation

Ex. 1: Evaluate $\lim_{x\to 0} \frac{e^x - 1}{\sin x}$ we must write $\rightarrow \lim_{x\to 0} e^x - 1 = 0$ & $\lim_{x\to 0} \sin x = 0$

Since direct substitution leads to indeterminate $\frac{0}{0}$, L'Hôpital applies!!

$$\lim_{x\to 0} \frac{e^x - 1}{\sin x} = \lim_{x\to 0} \frac{e^x}{\cos x} = \frac{e^0}{\cos 0} = \frac{1}{1} = \boxed{1}$$

Ex. 2: Evaluate the following limits:

a) $\lim_{x\to 4} \frac{x^2 - x - 12}{x^2 - 16}$

$\lim_{x\to 4} (x^2 - x - 12) = 0$

$\lim_{x\to 4} (x^2 - 16) = 0$

$\therefore$ L'Hôpital applies

$$= \lim_{x\to 4} \frac{2x - 1}{2x} = \boxed{\frac{7}{8}}$$

just like we got before

b) $\lim_{h\to 0} \frac{\cos(x+h) - \cos x}{h}$

$\lim_{h\to 0} [\cos(x+h) - \cos x] = 0$

$\lim_{h\to 0} h = 0$

$\therefore$ L'HôP applies

$$= \lim_{h\to 0} \frac{-\sin(x+h) \cdot 1 - 0}{1}$$

$$= \boxed{-\sin x}$$

just like we got before

c) $\lim_{x\to\pi} \frac{\cos x + \sin(2x) + 1}{x^2 - \pi^2}$ is $\rightarrow 0$, $\rightarrow 0$

(A) $\frac{1}{2\pi}$

(B) $\frac{1}{\pi}$

(C) 1

(D) nonexistent

$\therefore$ L'HôP

$$\lim_{x\to\pi} \frac{-\sin x + 2\cos 2x}{2x}$$

$$= \frac{2}{2\pi} = \boxed{\frac{1}{\pi}}$$

Limits to Infinity

Ex. 3: Evaluate the following limits:

a) $\lim\limits_{x\to-\infty} \frac{3x^2+7x-5}{5-4x^2}$

$\lim\limits_{x\to-\infty}(3x^2+7x-5)=\infty$ } indeterminate!

$\lim\limits_{x\to-\infty}(5-4x^2)=-\infty$

$=\lim\limits_{x\to-\infty}\frac{6x+7}{-8x}$ another round!!

$\lim\limits_{x\to-\infty}(6x+7)=-\infty$ } indeterminate!

$\lim\limits_{x\to-\infty}-8x=\infty$

$=\lim\limits_{x\to-\infty}\frac{6}{-8}=\boxed{-\frac{3}{4}}$

b) $\lim\limits_{t\to\infty} t^2e^{-2t}$

$\infty\cdot 0 \to$ also indeterminate!! But let's get a fraction!

$=\lim\limits_{t\to\infty}\frac{t^2}{e^{2t}}$

$\lim\limits_{t\to\infty}t^2=\infty,\ \lim\limits_{t\to\infty}e^{2t}=\infty$

$=\lim\limits_{t\to\infty}\frac{2t}{2e^{2t}}$

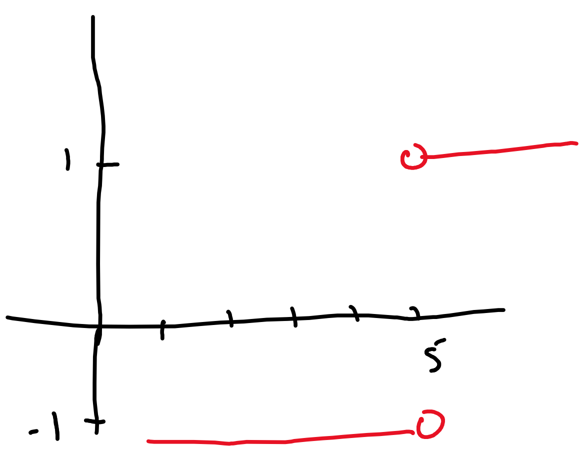
$\lim\limits_{t\to\infty}2t=\infty,\ \lim\limits_{t\to\infty}2e^{2t}=\infty$

$=\lim\limits_{t\to\infty}\frac{2}{4e^{2t}}=\boxed{0}$

Ex. 4: Evaluate the following limits:

a) $\lim\limits_{p\to5}\frac{|p-5|}{p-5}$

we don't know differentiating absolute value... but we need to know this graph!



$\boxed{\text{Limit DNE!}}$

b) $\lim\limits_{x\to\infty}\frac{\sqrt{x^2+5}}{x+1}$

$\lim\limits_{t\to\infty}\sqrt{x^2+5}=\infty$

$\lim\limits_{t\to\infty}(x+1)=\infty$

$\lim\limits_{x\to\infty}\frac{\frac{1}{2}(x^2+5)^{-1/2}\cdot 2x}{1}$

$=\lim\limits_{x\to\infty}\frac{x}{\sqrt{x^2+5}}$

$\lim\limits_{x\to\infty}x=\infty$

$\lim\limits_{x\to\infty}\sqrt{x^2+5}=\infty$

$=\lim\limits_{x\to\infty}\frac{1}{\frac{1}{2}(x^2+5)^{-1/2}\cdot 2x}$

$=\lim\limits_{x\to\infty}\frac{\sqrt{x^2+5}}{x}$ .... wait, we could do this forever...

$\therefore$ L'Hôp is useless

$\lim\limits_{x\to\infty}\frac{\sqrt{x^2}}{x}=\lim\limits_{x\to\infty}\frac{x}{x}=\boxed{1}$

Ex. 5: Evaluate the following limits without a calculator:

a) $\lim_{x\to\pi} \frac{\sin x}{x}$

$= \frac{0}{\pi}$

$= \boxed{0}$

b) $\lim_{x\to\pi} \frac{\sin x}{x-\pi}$

$\lim_{x\to\pi} \sin x = 0$

$\lim_{x\to\pi} (x-\pi) = 0$

$\therefore$ L'Hôp

$= \lim_{x\to\pi} \frac{\cos x}{1} = \boxed{-1}$

c) $\lim_{x\to\pi} \frac{\int_\pi^x \sin t\, dt}{x-\pi}$

$\lim_{x\to\pi} \int_\pi^x \sin t\, dt = 0$

$\lim_{x\to\pi} (x-\pi) = 0$

$= \lim_{x\to\pi} \frac{\sin x - 0}{1}$

$= \frac{0}{1} = \boxed{0}$

d) $\lim_{x\to 0} \frac{\cos^2 x-1}{e^{3x}-3x-1}$

$\lim_{x\to 0} \cos^2 x - 1 = 0$

$\lim_{x\to 0} (e^{3x} - 3x - 1) = 0$

$\therefore$ L'Hôp!

$= \lim_{x\to 0} \frac{-2\cos x \sin x}{3e^{3x} - 3}$

$\lim_{x\to 0} -2\cos x \sin x = 0$

$\lim_{x\to 0} 3e^{3x} - 3 = 0$

$\therefore$ Another L'Hôp!

$= \lim_{x\to 0} \frac{2\sin^2 x - 2\cos^2 x}{9e^{3x}} = \boxed{-2/9}$

e) $\lim_{x\to 0} \frac{7x-\sin x}{x^2+\sin(3x)} =$ $\to 0$ $\to 0$

(A) 6

(B) ~~2~~

(C) 1

(D) 0

$= \lim_{x\to 0} \frac{7-\cos x}{2x+3\cos 3x}$

$= \frac{6}{3}$

$= \boxed{2}$

f) $\lim_{x\to\infty} \frac{5x^{100}}{e^x}$

after about 100 rounds of L'Hôp....

$\lim_{x\to\infty} \frac{\text{constant}}{e^x}$

$= \boxed{0}$

g) If $f(x) = (x^3+1)^2$, find $\lim_{x\to 1} \frac{f(x)-f(1)}{x-1}$

$\lim_{x\to 1} \frac{(x^3+1)^2 - 4}{x-1}$

$\lim_{x\to 1} [(x^3+1)^2 - 4] = 0$

$\lim_{x\to 1} (x-1) = 0$

$\therefore$ L'Hôp!!

$= \lim_{x\to 1} \frac{2(x^3+1)\cdot 3x^2}{1} = \boxed{12}$

Note that g) is the alternate form of the derivative of f at x = 1.....

$f'(x) = 2(x^3+1)\cdot 3x^2$

$\boxed{f'(1) = 12}$

AP Calculus II
Notes 8.8
Improper Integrals

The definition of a definite integral $\int_a^b f(x)dx$ requires that the interval $[a, b]$ be finite. Also, the FTC requires that the function is continuous on $[a, b]$ in order to evaluate the antiderivative. Here, we will look at a procedure for evaluating integrals that does not satisfy any of these requirements. Integrals that have either one or both of the limits of integration which are infinite or have a finite number of discontinuities are known as **improper integrals**.

<u>Improper Integrals With Infinite Integration Limits</u>:

1. If f is continuous on the interval $[a, \infty)$, then $$\int_a^\infty f(x)dx = \lim_{b\to\infty}\int_a^b f(x)dx$$

2. If f is continuous on the interval $(-\infty, b]$, then $$\int_{-\infty}^b f(x)dx = \lim_{a\to-\infty}\int_a^b f(x)dx$$

3. If f is continuous on the interval $(-\infty, \infty)$, then $$\int_{-\infty}^\infty f(x)dx = \int_{-\infty}^c f(x)dx + \int_c^\infty f(x)dx$$

In the first two cases, the improper integral **converges** if the limit exists, otherwise the improper integral **diverges**. In the third case, if either of the two diverge, then the original integral diverges.

<u>Ex. 1:</u> Evaluate the following integral: $\int_1^\infty \frac{dx}{\sqrt[5]{x}} = \int_1^\infty x^{-1/5}dx$

$$\lim_{b\to\infty}\int_1^b x^{-1/5}dx = \lim_{b\to\infty} \frac{5}{4}x^{4/5}\Big|_1^b$$

$$= \lim_{b\to\infty}\left[\frac{5}{4}b^{4/5} - 5/4\right]$$

as $b\to\infty$ $\downarrow$ $b^{4/5}\to\infty$

$\infty - 5/4 \to \infty$ $\therefore$ $\boxed{\text{Diverges}}$

pick any constant

Ex. 2: Evaluate the following improper integrals:

a) $\displaystyle\int_1^{\infty} \frac{dx}{2x+1}$

$\lim_{b\to\infty} \frac{1}{2}\int_1^b \frac{2}{2x+1}\,dx$

$u = 2x+1,\ du = 2\,dx$

$\lim_{b\to\infty} \int_3^{\infty} \frac{1}{u}\,du$

$\lim_{b\to\infty} \ln|u|\Big|_3^b$

$\lim_{b\to\infty} [\ln b - \ln 3] \to \infty$

$\boxed{\text{Diverges}}$

b) $\displaystyle\int_{-\infty}^{\infty} \frac{dt}{4+t^2} = \int_{-\infty}^{0} \frac{1}{4+t^2}\,dt + \int_0^{\infty} \frac{1}{4+t^2}\,dt$

$= \lim_{a\to-\infty} \int_a^0 \frac{1}{4+t^2}\,dt + \lim_{b\to\infty}\int_0^b \frac{1}{4+t^2}\,dt$

$a^2 = 4 \quad u^2 = t^2$

$a = 2 \quad u = t \quad du = dt$

$\lim_{a\to-\infty} \frac{1}{2}\arctan\frac{t}{2}\Big|_a^0 +$

$\lim_{b\to\infty} \frac{1}{2}\arctan\frac{t}{2}\Big|_0^b$

$\lim_{a\to-\infty} \left(0 - \frac{1}{2}\arctan\frac{a}{2}\right) +$

$\lim_{b\to\infty} \left(\frac{1}{2}\arctan\frac{b}{2} - 0\right) = -\frac{1}{2}\left(-\frac{\pi}{2}\right) + \frac{1}{2}\left(\frac{\pi}{2}\right)$

$= \boxed{\pi/2}$

Ex. 3: Evaluate $\displaystyle\int_1^{\infty}\left(\frac{1}{p+4} - \frac{1}{p+6}\right)dp$

$u_1 = p+4 \qquad u_2 = p+6$

$du = 1\,dp \qquad du = 1\,dp$

$\lim_{b\to\infty} \int_1^b \left(\frac{du_1}{u_1} - \frac{du_2}{u_2}\right) = \lim_{b\to\infty}\left[\ln|p+4| - \ln|p+6|\Big|_1^b\right]$

$= \lim_{b\to\infty}\left[\underbrace{\ln|b+4| - \ln|b+6|}_{} - \ln 5 + \ln 7\right]$

$\ln\infty - \ln\infty \to \infty - \infty \Leftarrow$ INDETERMINATE!!!

we don't know which ∞ is greater

$= \lim_{b\to\infty}\left[\ln\left|\frac{b+4}{b+6}\right| - \ln 5 + \ln 7\right]$ as $b\to\infty$ $\frac{b+4}{b+6} \to 1$

$= \ln 1 - \ln 5 + \ln 7 = \boxed{\ln 7/5}$

Improper Integrals With Infinite Discontinuities:

1. If f is continuous on the interval $[a,b)$ and has a discontinuity at b, then $\int_a^b f(x)dx = \lim_{c\to b^-}\int_a^c f(x)dx$
2. If f is continuous on the interval $(a,b]$ and has a discontinuity at a, then $\int_a^b f(x)dx = \lim_{c\to a^+}\int_c^b f(x)dx$
3. If f is continuous on the interval $[a,b]$ except at c, then $\int_a^b f(x)dx = \int_a^c f(x)dx + \int_c^b f(x)dx$

In the first two cases, the improper integral **converges** if the limit exists, otherwise the improper integral **diverges**. In the third case, if either of the two diverge, then the original integral diverges.

Ex. 4: Evaluate the following improper integrals:

a) $\int_0^2 \frac{dn}{(n-2)^3}$

n=2 is domain restriction

Since we integrate from n=0 to n=2, we must approach 2 from left side

$$\lim_{b\to 2^-}\int_0^b \frac{1}{(n-2)^3}dn$$

$u = n-2$
$du = 1\,dn$

$$\lim_{b\to 2^-}\int_{-2}^{b-2} u^{-3}\,du \qquad \frac{u^{-2}}{-2}$$

$$\lim_{b\to 2^-}\left[\frac{-1}{2u^2}\Big|_{-2}^{b-2}\right]$$

$$\lim_{b\to 2^-}\left[\frac{-1}{2(b-2)^2}+\frac{1}{8}\right]$$

as $b\to 2^-$, $(b-2)^2 \to 0$

So, $\frac{-1}{0}\to\infty$ .... So $\infty + 1/8$ $\boxed{\text{Diverges}}$

b) $\int_{-1}^8 \frac{dx}{\sqrt[3]{x}}$

Domain restriction at $x=0$... Must split up!

$$\int_{-1}^0 x^{-1/3}dx + \int_0^8 x^{-1/3}dx$$

$$\lim_{b\to 0^-}\int_{-1}^b x^{-1/3}dx + \lim_{a\to 0^+}\int_a^8 x^{-1/3}dx$$

$$\lim_{b\to 0^-}\frac{3}{2}x^{2/3}\Big|_{-1}^b + \lim_{a\to 0^+}\frac{3}{2}x^{2/3}\Big|_a^8$$

$$=\lim_{b\to 0^-}\left[\frac{3}{2}b^{2/3}-\frac{3}{2}\right]+\lim_{a\to 0^+}\left[6-\frac{3}{2}a^{2/3}\right]$$

as $a\to 0^+$ and $b\to 0^-$

$\frac{3}{2}b^{2/3}\to 0$

$$=0-\frac{3}{2}+6-0$$

$$=\boxed{9/2}$$

Ex. 5: A solid is formed by revolving, around the x −axis, the unbounded region lying between the x −axis and the graph $f(x) = \begin{cases} 6-x & 1 \le x \le 4 \\ \sqrt{xe^{4-x}} & x > 4 \end{cases}$. Find the volume of the solid formed given $x \ge 1$.

$V = \pi \int_a^b (R(x))^2 dx$ ... but it's piecewise, so we need 2 integrals!

improper integral!

$$\pi \int_1^4 (6-x-0)^2 dx + \pi \int_4^\infty \left(\sqrt{xe^{4-x}} - 0\right)^2 dx$$

$R(x) =$ curve − axis

$R(x) =$ curve − axis

$$\pi \int_1^4 (6-x)^2 dx + \lim_{b\to\infty} \pi \int_4^b xe^{4-x} dx$$

$u = 4-x$, $du = -1dx$

No ....

Integration by parts!

$u = 6-x$, $du = -1dx$

$u = x$
$du = 1dx$

$\int dv = \int e^{4-x}$
$v = -e^{4-x}$

$u = 4-x$
$du = -1dx$

$$-\pi \int_5^2 u^2 du = \pi \int_2^5 u^2 du$$

$$\pi \int_2^5 u^2 du + \lim_{b\to\infty} \pi \left[-xe^{4-x} + -\int_4^b -e^{4-x} dx\right]$$

$u = 4-x$
$du = -1dx$

$$\pi \left[\frac{1}{3}u^3 \Big|_2^5\right] + \lim_{b\to\infty} \pi \left[-xe^{4-x} - e^{4-x} \Big|_4^b\right]$$

$-e^{-\infty} \to 0$

$$\pi \left[\frac{125}{3} - \frac{8}{3}\right] + \lim_{b\to\infty} \pi \left[\underbrace{-be^{4-b}} - \underbrace{e^{4-b}} + 4 + 1\right]$$

$-\infty \cdot 0$ is indeterminate....

so L'Hôp!!

$$= 39\pi + 5\pi + \lim_{b\to\infty} \frac{-\pi b}{e^{b-4}} = 44\pi + \lim_{b\to\infty} \frac{-\pi}{e^{b-4}} =$$

$\lim_{b\to\infty} -\pi b \to \infty$, $\lim_{b\to\infty} e^{b-4} \to \infty$ $\therefore$ L'Hôp applies!!

$\boxed{44\pi + 0}$

Ex. 6:

x	1	2	3	4	5
$f'(x)$	62	30	20	15	12

The function f is twice differentiable for $x \geq 1$. Selected values of the positive and decreasing function f' are given in the table above. The graph of f' has a horizontal asymptote at $y = 0$.

a) Evaluate $\int_1^{\infty} f''(x)\sin(f'(x))\,dx$.

$u = f'(x) \quad du = f''(x)\,dx$

$$\lim_{b\to\infty}\int_{f'(1)}^{f'(b)} \sin u\,du = \lim_{b\to\infty}\int_{62}^{f'(b)} \sin u\,du$$

$$= \lim_{b\to\infty}\left[-\cos u\Big|_{62}^{f'(b)}\right] = \lim_{b\to\infty}\left[-\cos(f'(b)) + \cos 62\right]$$

* a horizontal asymptote of f' means $\lim_{b\to\infty} f'(b) = 0$ *

$$= -\cos 0 + \cos 62 = \boxed{\cos 62 - 1}$$

b) Evaluate $\int_1^5 \left(\frac{f''(x)}{\sqrt{f'(x) - 12}}\right) dx$.

there's a domain restriction if $f'(x) - 12 = 0$ or when $f'(x) = 12$. This happens at $x = 5$ ... so IMPROPER!!!

$$\lim_{b\to 5^-}\int_1^b \frac{f''(x)}{\sqrt{f'(x)-12}}\,dx$$

$u = f'(x) - 12 \qquad x = 1 \to u = 50$

$du = f''(x)\,dx \qquad x = b \to u = f'(b) - 12$

$$\lim_{b\to 5^-}\int_{50}^{f'(b)-12} u^{-1/2}\,du = \lim_{b\to 5^-} 2\sqrt{u}\Big|_{50}^{f'(b)-12}$$

$$= \lim_{b\to 5^-}\left[\underbrace{2\sqrt{f'(b)-12}}_{\text{as } b\to 5,\ f'(b)\to 12} - 2\sqrt{50}\right] = 0 - 2\sqrt{50}$$

$$= \boxed{-2\sqrt{50}}$$